VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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Synopsis Report on

"Fire Detector IoT System with Extinguisher Class Recommendation"

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Submitted by

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ABSTRACT

Fires cause significant harm by endangering lives, destroying property, polluting the air, and leading to economic losses. Early detection and prompt response can help mitigate these risks. This project presents an Internet of Things (IoT) based fire detection system that automatically identifies fires, determines the type of burning object, recommends the appropriate fire extinguisher class, and sends real-time notifications with location details to users and emergency responders. The system detects fire through advanced sensors and analyzes the affected object using a deep-learning-based algorithm. Once a fire is detected, the system transmits data to a cloud-based server, which then relays alerts to connected smartphone applications. This innovation fills a critical gap by not only detecting fire but also providing extinguisher recommendations—helping firefighters respond swiftly and effectively. Prototypes of the fire detection device, central emergency server, and mobile applications have been developed and tested successfully.

INTRODUCTION

Fire hazards pose a significant threat to life, property, and the environment, making early detection and appropriate response crucial. Traditional fire detection systems often rely on basic smoke or heat sensors, which may lead to delayed responses and false alarms. To address these limitations, this project proposes a Smart Fire Detector IoT System with Extinguisher Class Recommendation, integrating advanced sensors, data analysis, and cloud-based connectivity.

This system leverages IoT technology to detect fire incidents accurately by analyzing multiple parameters, including temperature, smoke density, and flame detection. Unlike conventional systems, it incorporates object identification to recognize the specific item that has caught fire. By utilizing machine learning and image processing, the system can determine the burning material and suggest the appropriate fire extinguisher class (A, B, C, D, or K) to suppress the fire effectively. A key feature of this system is its cloud-based architecture, allowing real-time monitoring and alerts through a web or mobile application. When a fire is detected, the system sends immediate notifications to emergency responders and nearby personnel, ensuring quick action.

OBJECTIVE

The objective of this project is:

- **IoT-based Fire Detection:** Develop a system that utilizes IoT technology to continuously monitor and detect fire incidents in real-time.
- **Extinguisher Class Recommendation:** Integrate intelligent algorithms to analyze the incident and recommend the appropriate fire extinguisher class (A, B, C, D, or K).
- **Rapid Alerts:** Ensure immediate notification and alerting to enable quick response and enhance overall fire safety.

PROBLEM STATEMENT

Traditional fire detection systems primarily rely on smoke and heat sensors, which may not always provide timely alerts, leading to increased damage and slower response times. Additionally, many existing systems lack the capability to identify the specific object that has caught fire, making it difficult to determine the appropriate firefighting approach.

There is a need for a reliable, real-time fire detection system that not only detects fire but also identifies the correct fire extinguisher class (A, B, C, D, or K).

EXISTING SYSTEM

Several systems have been developed for fire detection and management, incorporating IoT sensors, fire detection algorithms, and communication technologies. These systems are designed to monitor environmental parameters such as smoke, temperature, and gas levels to detect the presence of fire. However, most of these systems have limitations in intelligent decision-making, extinguisher recommendation, and real-time communication. Below is an overview of some key existing systems and their limitations:

- **Lack of Intelligent Decision-Making**

  Existing fire detection systems mainly use basic sensors to identify the presence of a fire but lack the ability to analyze the material. Without intelligent algorithms or machine learning, they cannot distinguish between different object is on fire. This absence of decision-making capability leaves emergency responders without crucial information, increasing the likelihood of using inappropriate suppression methods, which can worsen the situation.

- **Inability to Recommend Suitable Extinguishers**

  Different fire types require specific extinguisher classes for effective suppression, but most existing systems do not provide automated guidance on the correct extinguisher to use. Responders often have to rely on their judgment or external guidance, which can lead to the use of the wrong extinguishing agent. This not only increases the risk of further damage but can also result in hazardous reactions, particularly in cases involving electrical or chemical fires.

- **Limited Communication Methods**

  Current fire detection systems typically rely on basic alarms, sirens, or visual alerts to notify occupants of a fire. While some may send notifications to facility managers through mobile apps, they often lack detailed information such as fire location, severity, or recommended actions. Without real-time, actionable insights, emergency responders face delays in assessing the situation and making informed decisions, ultimately reducing the efficiency of the firefighting process.

PROPOSED SYSTEM

The proposed system offers an intelligent and integrated solution for fire detection and management using IoT sensors, machine learning algorithms, and real-time communication technologies. Unlike traditional fire detection systems, this solution not only detects fires but also classifies the type of fire and recommends the appropriate class of extinguisher (Class A, B, C, D, or K). Additionally, it ensures rapid emergency response through real-time notifications to relevant personnel.

1. **IoT Sensor-Based Fire Detection**
   - The system utilizes various IoT sensors, including smoke sensors, temperature sensors, flame detectors, and gas sensors, to continuously monitor environmental parameters.
   - Sensor data is transmitted to a centralized processing unit for real-time analysis, where anomalies indicating a fire are detected.
2. **Extinguisher Class Recommendation**
   - Using pre-trained machine learning models, it identifies the appropriate extinguisher class:
     - Class A: Ordinary combustibles like wood, paper, or cloth.
     - Class B: Flammable liquids such as gasoline, oil, or solvents.
     - Class C: Electrical equipment fires.
     - Class D: Combustible metals like magnesium or aluminum.
     - Class K: Kitchen fires involving cooking oils and fats.
   - The system then provides a real-time recommendation for the suitable extinguisher type, reducing response time and improving firefighting efficiency.
3. **Real-Time Notification and Alerting System**
   - In the event of a fire, the system generates immediate notifications via SMS, email, or mobile app alerts.
   - Alerts include critical information such as fire location, severity, sensor data, and the recommended extinguisher class.
   - Notifications are sent to emergency responders for efficient response.

LITERATURE SURVEY

[1] IoT Based Smart Home and Office Fire Notification Alert System

• Authors: Sunandita Debnath, Jashwin Gautam, Sahil Kumar Rai

• Conference: 2024 IEEE International Conference on Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI)

• Summary: Employs a network of temperature, smoke, and CO sensors integrated with an Arduino UNO that communicates via MQTT with a cloud platform to trigger alarms, sprinkler activation, and notifications upon abnormal readings. Provides real-time monitoring, remote control via a mobile application, and demonstrates high accuracy and responsiveness in extensive hardware testing. Relies on fixed threshold values that may not adapt well to varying conditions and lacks advanced machine learning for improved detection.

[2]A Smart Fire Detector IoT System with Extinguisher Class Recommendation Using Deep Learning

• Authors: Tareq Khan

• Journal: IoT, 2023, Volume 4, Pages 558–581 (DOI: 10.3390/iot4040024)

• Summary: Utilizes thermal and RGB cameras with the SSD-Inception-V2 deep learning model and homography transformation to detect fires and recommend the correct extinguisher class based on a lookup table. Enhances detection accuracy and provides tailored extinguisher recommendations, enabling rapid alerts and improved safety in indoor environments. Involves complex hardware and computationally intensive deep learning models that may increase costs and present alignment challenges.

[3] Forest Fire Detection and Notification Method Based on AI and IoT Approaches

• Authors: Kuldoshbay Avazov, An Eui Hyun, Alabdulwahab Abrar Sami, Azizbek Khaitov, Akmalbek Bobomirzaevich Abdusalomov, and Young Im Cho

• Journal: Future Internet, 2023, 15(2), 61 (DOI: 10.3390/fi15020061O)

• Summary: Deploys IoT sensors to continuously monitor environmental parameters and applies deep learning algorithms to detect forest fires, subsequently notifying authorities via SMS and other communication channels. Improves detection accuracy, reduces false alarms, and facilitates rapid response in forested areas. Faces potential connectivity issues in remote regions and requires extensive, diverse training data for robust deep learning performance.

[4] A Deep Learning Based Object Identification System for Forest Fire Detection

• Authors: Federico Guede-Fernández, Leonardo Martins, Rui Valente de Almeida, Hugo Gamboa, and Pedro Vieira

• Journal: Fire, 2021, 4(4), 75

• Summary: Leverages convolutional neural networks to analyze forest imagery for automated object identification associated with fire incidents. Significantly enhances early detection accuracy and reduces false positives, thereby enabling prompt emergency responses. Its performance can be affected by environmental conditions and the high computational demands may limit real-time processing on low-power edge devices.

SOFTWARE REQUIREMENTS

- **Language** : Python , Arduino, C++
- **Web Technologies** : HTML, CSS, Javascript, Django, Django Rest Framework, React,ChartJs, Geolocation API, Fetch API, ReactNative, Kaggle
- **Libraries** : Scikitlearn, Tensorflow

HARDWARE REQUIREMENTS

- Sensors(Temperature, Smoke)
- ESP-32
- GSM Module
- 12V or 5V DC Adapter

REFERENCES

[1] S. Debnath, J. Gautam, and S. K. Rai "IoT Based Smart Home and Office Fire Notification Alert System" **,** IEEE International Conference on Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI) 2024.

[2] Tareq Khan , "A Smart Fire Detector IoT System with Extinguisher Class Recommendation Using Deep Learning", IoT, Volume 4, Pages 558–581, 2023.

[3] K. Avazov, A. E. Hyun, A. A. Abrar Sami, A. Khaitov, A. B. Abdusalomov, and Y. I. Cho "Forest Fire Detection and Notification Method Based on AI and IoT Approaches", Future Internet, 2023.

[4]F. Guede-Fernández, L. Martins, R. V. de Almeida, H. Gamboa, and P. Vieira. "A Deep Learning Based Object Identification System for Forest Fire Detection", 2024.

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